Bond Risk Lab

Convexity Sensitivity AI Agent

Professional Submission Documentation Structure

Prepared for the Zetheta Data Analyst - Convexity Sensitivity AI Agent project. This document provides the exact report structure, deliverable arrangement, testing checklist, and submission mapping for the final project package.

Project Category	Quantitative Finance | Fixed Income Analytics | Machine Learning
Core Platform	Streamlit Bond Risk Lab simulation app
Analytics Scope	Duration, Convexity, DV01/PVBP, KRD, Yield Curve, Monte Carlo VaR, ML
Submission Assets	Python app, dataset, Power BI DAX, R script, Excel/VBA notes, PDF structure

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1. Executive Summary

State the objective: building an AI-powered fixed income risk platform that measures portfolio sensitivity under changing market conditions. Include 300-bond portfolio scope, market value, duration, convexity, DV01, and rate-shock P&L.; End with one paragraph explaining how the app trains analysts through the gamified Bond Risk Lab.

2. Business Problem and Use Case

Describe the CIO requirement: stress-test a multi-currency bond portfolio under interest-rate shocks, yield-curve moves, and Monte Carlo scenarios. Explain the value for treasury desks, ALM teams, pension funds, insurance companies, and fixed income risk analysts.

3. Dataset and Portfolio Design

Document columns in data/bond_portfolio.csv: BondID, InstrumentType, Currency, CreditRating, CouponRate, YieldToMaturity, YearsToMaturity, CleanPrice, MarketValue_INR, ModifiedDuration, EffectiveDuration, Convexity, DV01_Per100Face, Quantity, TenorBucket. Include data dictionary and portfolio segmentation.

4. Quantitative Methodology

Explain bond price, Macaulay Duration, Modified Duration, Effective Duration, Convexity, DV01/PVBP, Key Rate Duration, and duration-convexity price impact. Include the formula: Delta P/P approx -Modified Duration x Delta y + 0.5 x Convexity x Delta y^2.

Measure	Purpose
Modified Duration	Linear price sensitivity to yield movements
Convexity	Second-order curvature correction
DV01/PVBP	Absolute INR value of a one basis point move
KRD	Sensitivity by tenor bucket across the yield curve

5. Yield Curve Scenario Lab

Explain parallel, steepening/flattening, and butterfly shocks. Mention the app sliders for yield curve scenario generation and how the scenario curve visually compares to the base curve.

6. Monte Carlo VaR and Expected Shortfall

Explain simulation of interest-rate paths, distribution of portfolio P&L;, VaR 95%, and CVaR 95%. Mention the Monte Carlo tab and histogram visualization.

7. Machine Learning Agent

Describe the Random Forest price prediction model using coupon, maturity, yield, duration, convexity, rating, and tenor bucket. Mention RMSE, R2, and feature importance chart. Optional future upgrade: XGBoost and neural network comparison.

8. Gamified Bond Risk Lab

Explain daily analyst challenge, hedge decision selection, scoring rules, and instant feedback. Describe the smoother single-section rendering and animated option cards added for the final version.

9. Power BI, R, and Excel/VBA Deliverables

Reference powerbi/BOND_RISK_DAX_MEASURES.txt, r/bond_risk_models.R, and excel/BOND_RISK_EXCEL_VBA.txt. Explain how these satisfy the multi-platform implementation requirement.

10. Testing and Validation

Include checks: app launches from app.py, filters work, charts render, no duplicate Plotly ID errors, selection process is fast, Monte Carlo runs, ML tab degrades gracefully if scikit-learn is unavailable, download buttons work.

Test Item	Expected Result
Launch command	python -m streamlit run app.py opens dashboard
Navigation	Only selected section renders, improving speed
Risk Game	Option selection is responsive and feedback appears
Charts	Plotly charts render without duplicate-element error
Submission Pack	Checklist and DAX downloads available

11. Final Submission Arrangement

Submit the complete ZIP with app/, src/, data/, docs/, powerbi/, r/, excel/, outputs/, requirements.txt, app.py, run_app_windows.bat, run_bond_risk_lab_windows.bat, README.md, and checklist files. Keep the folder name short and avoid nested duplicate folders.

12. Future Enhancement Roadmap

List future upgrades: XGBoost model, neural network model, live yield curve API integration, SHAP explainability, downloadable PDF report generation from app, Streamlit Cloud deployment, and Power BI PBIX dashboard.

Submission Folder Checklist

Folder/File	Submit?	Purpose
app.py	Yes	Root launcher for Streamlit
app/bond_risk_lab_app.py	Yes	Main dashboard application
src/bond_risk_engine.py	Yes	Quant engine
data/bond_portfolio.csv	Yes	Bond portfolio dataset
powerbi/BOND_RISK_DAX_MEASURES.txt	Yes	Power BI measures
r/bond_risk_models.R	Yes	R implementation
excel/BOND_RISK_EXCEL_VBA.txt	Yes	Excel/VBA implementation notes
docs/Bond_Risk_Lab_Submission_Documentation_Structure.pdf	Yes	PDF structure/documentation
requirements.txt	Yes	Dependencies
README.md	Yes	Run instructions